CS 431 Operating Systems Analysis/Design

Final Project Description
(Analysis of an Operating System)

Context

Your goal in this course is to learn the basics of operating systems for centralized computers in general, and to learn to evaluate the appropriateness of an operating system for a particular environment -- not to learn the details of a particular operating system.

Task Description

To provide a general survey of operating systems, perform an in-depth analysis of an operating system as your Final Project. Start working on your Final Project (Analysis of an Operating System), which is due in **Week Seven**.

Deliverable

Prepare an analysis of your chosen operating system. The written analysis is expected to be 11-15 pages of text (meaning the page count does ***not*** include any graphics or illustrations, the title page, the table of contents or the reference list).

Please refer to the Characteristics of Effective Writing section at the end of this document for a quick reference guide as you write your analysis.

The following topics are **required** in your analysis of the operating system:

- The design principles underlying the operating system (batch, on-line, etc.).
- An overview of all the major elements/components.
- The advantages and disadvantages of this operating system (in which environments would this operating system work best).

You are also **required** to include coverage of **3 of the following 5** topics (*your choice*):

- The major elements of process management (process status, process control, synchronization, concurrency control, etc.).
- The methods used for inter-process communication (shared memory or message systems, direct or indirect, one-way or two-way, e.g. mailboxes, pipes or ports) and how the OS handles I/O functions.
- The major elements of memory management (overlays, partitions, multi-programming, virtual memory, thrashing, etc.).
- The major elements of scheduling (algorithms used, deadlock handling, potential starvation problems, etc.).
- The major elements of file system management (access methods, directory systems, file protection, etc.).

The following additional topics ***may optionally*** also be included in your analysis of the operating system:

- The major elements of the programmer interface (Job Control Language, what the programmer needs to know to use the system, etc.).
- The major elements of the user interface (Job Control Language, what the user needs to know to use the system, etc.).
- Networking capabilities of the operating system.
- Distributed and /or virtual capabilities of the operating system.

In order for you to pace yourself for this important project, the weekly tasks are provided below. If you have any questions, please contact your facilitator. You may use the weekly Question & Answer Discussion Forums, or if private, use Course Mail.

Assessment/Grading Criteria
Refer to the Final Project Rubric located in the Course Resources folder for this activity's grading criteria.

What to do each week

Week 1

Week 2

Week 3

Week 4

Week 5

Week 6

Week 7

Week 8

Characteristics of Effective Writing

Week One: Select Your Top Two

Select your top two computer operating systems that you would like to research for your Final Project. You may consider a microcomputer operating system only if it incorporates mainframe features such as virtual memory, multi-tasking, and network communications, etc. Below are some possible operating systems to consider for your selection (or you may propose your own).

Note: *To ensure that the major operating systems are researched and discussed in this class, the facilitator may assign an operating system different from the one(s) you selected.

- IBM: MVS, OS/360, OS/390, OS/400, z/OS, z/VM
- Windows: NT, XP, Vista, Windows 7, Windows 8
- Apple: Classic Mac OS, Rhapsody, NeXTstep, OS X, iOS
- Proprietary UNIX: A/UX, HP-UX, Solaris, IRIX, OSF/1, Ultrix, System V, etc.
- Open Source UNIX: BSD, Linux, Ubuntu, SCO, Open Solaris, etc.
- Research (Unix-like): Amoeba, Minix
- Symbian OS
- DEC VMS
- Mach
- AtheOS
- Realtime: VXWorks, velOSity, AMX RTOS
- Google Chrome OS
- Android

Deliverable

By 11:59 PM MST Sunday of Week One, post your top **two** selections for your Final Project research of an operating system, in order of preference to the Discussion Forum titled for this activity. Your facilitator will let you know which of your operating system selections you will cover, or will provide you with another option.*

Weeks Two and Three: Research for Final Project & Discussion

The focus of the research for you Final Project is a review of recent articles, within the field of operating system design, which provides you with the breadth and depth knowledge of your chosen operating system. It is important for you to be able to research and discuss the general structure, design directions, and theory based on current research and literature.

Task Description

Note: You will use your research from Weeks Two and Three discussion assignments in order to prepare to draft your Final Project Outline due in Week Three.

Your initial response must be posted to the Discussion Forum titled for this activity by 11:59 pm MST Thursday of Weeks Two and Three respectively.

Step 1: To begin your research for your Final Project's selected operating system, select a glossary term from your assigned textbook readings.

Note: For those who have access to the technical documentation set for an operating system, note that in an eight week course such as this one, it can be difficult to pick out the relatively high level information for your Final Project from a set of 12 to 24 detailed technical volumes. It is generally easier to use overview articles from the IEEE, ACM, NASPA, DECUS, etc. professional society publications. These publications are available throughout local college and public libraries.

Additionally, you will want to research through Regis' online library databases available to you for free. If you are not familiar with the library search please review the tutorial, Regis Library Computer Science Research Tutorial. You will need to hover your mouse over this title for the link to appear. If you prefer, select the print version (pdf) of the tutorial by selecting, [PDF Alternative].

In general, the following will assist you in accessing and researching the many free resources Regis Library resources available to you:

1. Open a browser and go to Regis Library homepage, http://www.regis.edu/library.htm.
2. In the left column, select the "Research by Subject" option.
3. Select the "Computer & Information Science" option.
4. You will see that you have many Popular Resources to choose from.
5. Select a resource/database to explore. If you are prompted, use your RegisNet username and id and password, the same credentials you used to log into this course.
6. If prompted to proceed to Outside Service (after logging in), do so.
7. You should see search options. Since each resource/database search options vary, you will need to explore.

Note: Many times your searches may provide you with the abstract of an article, and NOT the full article. You may need to look all over the page to locate a link to the full article.

Step 2: Next, using the same glossary term from Step 1 above, conduct a Google search for relevant articles.

Step 3: Compare your findings from both of your search results. Present your research findings in two or more paragraphs addressing each of the following. Provide a rationale for your responses and cite your sources and any relevant theories, concepts, protocols, and/or

standards that you used in your response. Be sure to use American Psychological Association (APA) style and format for all of your written work.

1. What was the research problem discussed in the article(s)?
2. What were the research findings?
3. What is the relevance and value of the article(s) for your Final Project?

Note: Looking ahead you will need to provide your list of resources as References in APA style and format as part of your initial Final Project Outline next week.

Step 4: Further, respond by **11:59 pm MST Sunday of Weeks Two and Three, respectively,** to two or more of your colleagues' postings in one or more of the following ways:

- Ask a probing question.
- Share an insight from having read a colleague's posting.
- Offer and support an opinion.
- Validate an idea with your own experience.
- Make a suggestion.
- Expand on a colleague's posting.

Check back throughout the week to further engage in the discussion.

Deliverable
Post your response(s) to the Discussion Forum titled for this activity by the dates specified.

Assessment/Grading Criteria
Refer to the Writing and Discussion Rubrics located in the Course Resources folder for this activity's grading criteria.

Week Three: Post an Outline

Prepare a brief outline with a short description about your Final Project that includes but not limited to:

- the operating system that you will discuss (approved by facilitator)
- why you selected that operating system
- what will you cover in your Final Project about your selected operating system
- your initial list of resources from your research about your selected operating system. Resources and citations must be in APA style and format.

Deliverable
Submit your Final Project Outline in the course Drop Box **by 11:59 pm MST Sunday of Week Three** to receive facilitator feedback.

Weeks Four and Five: Begin Drafting

Final Project
Build upon your outline from last week by drafting your Final Project.

Deliverable None

Week Six: Revise and Edit
Revise your draft if needed. If you have any questions for your facilitator, post those to the Discussion Forum.

Deliverable None

Week Seven: Polish and Post
Final Project - Polish and Post to the Week Eight discussion forum titled, Final Project.
To post your Final Project, you will need to **compose a new message** with the name of your operating system and your name in the subject line. For example, Subject: UNIX by Joe Smith.

Deliverable
1. By 11:59 PM MST **Sunday of Week Seven**, post to the ***Week Eight Discussion Forum.***
2. Attach your Final Project file.

Assessment/Grading Criteria
Refer to the Final Project Rubric located in the Course Resources folder for this activity's grading criteria.

Week Eight: Feedback & Discussion

Review at least two of your peer's Final Project papers, and provide a response to their operating system project and research findings.

Step 1: Please try to be thorough, to make your responses worthwhile for everyone! Please note that your responses and comments must have substance. For example, a response or comment of "good job" etc., are NOT acceptable. Responses for each of your two or more peer's Final Projects should be a ***MINIMUM of two paragraphs***.

Important Note: Your initial responses must be a Reply to the two or more peer's Final Projects **by 11:59 pm MST Thursday of Week Eight** to receive credit.

Your response should include:

- Thoughts on the overall quality of the project.
- Notes on features of the operating system which surprised you.
- Further expansion on any feature you have knowledge about.
- A comparison of the operating system to the one your Final Project was about.
- Anything else about the operating system and/or final project that you think would be of interest to the class.

Step 2: Check back throughout the week to further engage in the discussion.

Deliverable

Post your responses to the Discussion Forum titled for this activity by the

Assessment/Grading Criteria

Refer to the Discussion Rubric located in the Course Resources folder for this activity's grading criteria.

Characteristics of Effective Writing

Main Idea

The paper is focused, meets the expectations set up by the writer, and makes these expectations clear to the reader. The paper shows a clear sense of purpose.

Well-Organized

The paper is clearly developed; transitions are clear from sentence to sentence and paragraph to paragraph. In other words, the writer has not simply made a series of unrelated or vaguely related statements. Rather, each sentence and paragraph carries the reader closer to an understanding or appreciation of the writer's goal.

Support

The writer provides specific, concrete and appropriate information from memory, observation, reading, interviewing or other sources. The paper is well developed with examples, details, illustrations, anecdotes or the like.

Style

Sentences are varied and word choices are accurate. There is an absence of "clutter" or "padding". Phrasing is clear and direct. Tone is handled consistently; sentence length and word choice are appropriate to the audience and purpose of the piece.

Mechanics

Punctuation, grammar, spelling and aspects of format are handled correctly. The writer has prepared the paper carefully with attention to appearance and other details. Opening, closing and title are strong and contribute to the sense of purpose, focus and unity of the piece of writing.

Proofreading

Ask yourself the following questions:

- What is my major point?
- Have I supported generalizations, opinions and conclusions with specific examples?
- Have I avoided using unnecessary words and clichés?
- Have I proofread for spelling, punctuation and typographical errors?